The Student-Powered SOC Handbook

Protect your university today while training the security leaders of tomorrow

splunk>
a CISCO company

We all had our favorite corner, cubby, or carrel at our college library.

Was it tucked behind a long-neglected taxidermied exhibit on the sharp-tailed grouse? Was it in a cinder-block basement, awash in a vending machine's soft, promising glow? Or, perhaps you preferred to work back at the dorm, where frequent knocks at the door reminded you that you'd been studying for long enough already! Put down the books. Put on your toga.

Wherever you chose to cram for finals or knock out a late-night essay, the security of your data was probably the last thing on your mind. Good.

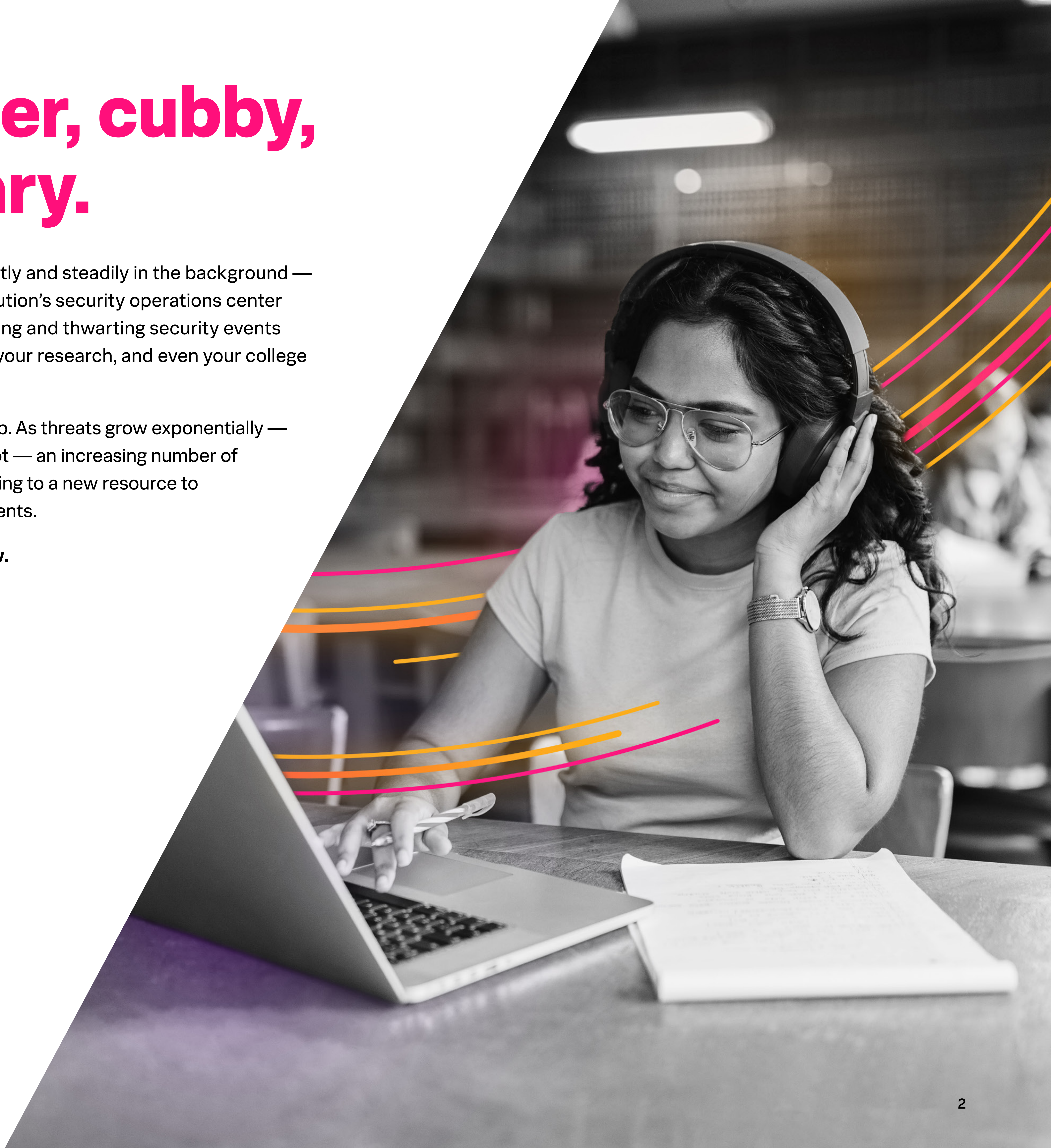
That's because — working quietly and steadily in the background — the security team in your institution's security operations center (SOC) was hard at work detecting and thwarting security events that could have put your data, your research, and even your college career at risk.

But it's getting harder to keep up. As threats grow exponentially — while budgets and talents do not — an increasing number of higher education SOCs are turning to a new resource to bolster security: their own students.

This handbook shows you how.

I know the number of threats against the university is higher and the complexity greater than ever before. But I also know that we're protecting the university at a level we've never had before.

— Bill Britton, Chief Information Officer, Cal Poly



Securing our ivory towers: Universities and growing cyber threats

Universities are notorious for being vulnerable to phishing attacks. California Polytechnic State University (Cal Poly), for one, receives over a million hits a day that they consider attacks on the university. But the security challenges higher education institutions face extend far beyond one vector, and it's complicated by a number of factors — from the complexity of cyber attacks and IT environments, to the small and shrinking security workforce.

The volume and complexity of cyber attacks continue to be on the rise.

Higher education is increasingly a cybercrime target. As a whole, the education sector has seen a 75% year-over-year increase to 3,574 weekly attacks. The volume of attacks isn't the only issue, as bad actors continue to get smarter at exploiting the vast stores of personal information schools store and manage. An implication is the increased time needed to detect and remediate these pernicious attacks.

Complex hybrid environments expand attack surfaces.

Increasingly, security teams are tasked with wrangling more complex IT infrastructures and operating models as the IT service landscape becomes more complex. With sprawling, interdependent hybrid and multicloud technology stacks — which often rely on third-party services — higher education institutions are left with more services to monitor and secure, and more potential points of failure.

Colleges and universities struggle to fill IT staffing shortages.

Like many U.S. industries, the public sector faces a shrinking cybersecurity workforce. Nearly three-quarters (74%) of campus leaders say that hiring IT employees is a "moderate" or "severe" challenge. With only 69 trained workers available for every 100 jobs across the industry as a whole, many of those positions won't be filled, especially in a sector that struggles to compete with the eye-popping salaries and benefit packages of industries like technology and financial services.

The education sector has seen a 75% year-over-year increase to 3,574 weekly attacks.

Fulfilling two missions with one student-powered SOC

To address these challenges, some of the country's best institutions of higher learning are stepping up in an inventive way. They're tapping into the expertise and potential of their own students by welcoming them into the heart of their security operations through on-campus student-powered SOCs.

The core mission of every college and university is to teach and prepare students for a career or next chapter. By hiring students to work alongside professional security professionals, institutions offer students the opportunity to develop skillsets they can carry with them throughout their careers, while receiving real-world training that will set them up for future success.

And then, of course, is the core mission of an institution's security team: to protect its students and employees from bad actors trying to break in and do meaningful harm. Having students in the SOC multiplies the security team, giving professional staff the time and capacity to address strategic and operational things while the students "provide the first, tier-one response," says Doug Lomsdalen, chief information security officer for Cal Poly and one of the original leaders on the idea of the student-powered SOC concept.

At LSU, if an incident happened after six, seven, or eight o'clock at night, it would not get looked at until the next morning. Now, with the SOC program, we have 24/7 coverage, 365 days a year, increasing the overall security posture of LSU and the state.

— Sumit Jain, CISO, Louisiana State University

About the Splunk Academic Alliance

This is where we get to the fun part. To help you staff your own student-powered SOC, we offer Splunk training and industry-recognized Splunk certifications at no cost to university students, faculty, and IT staff.

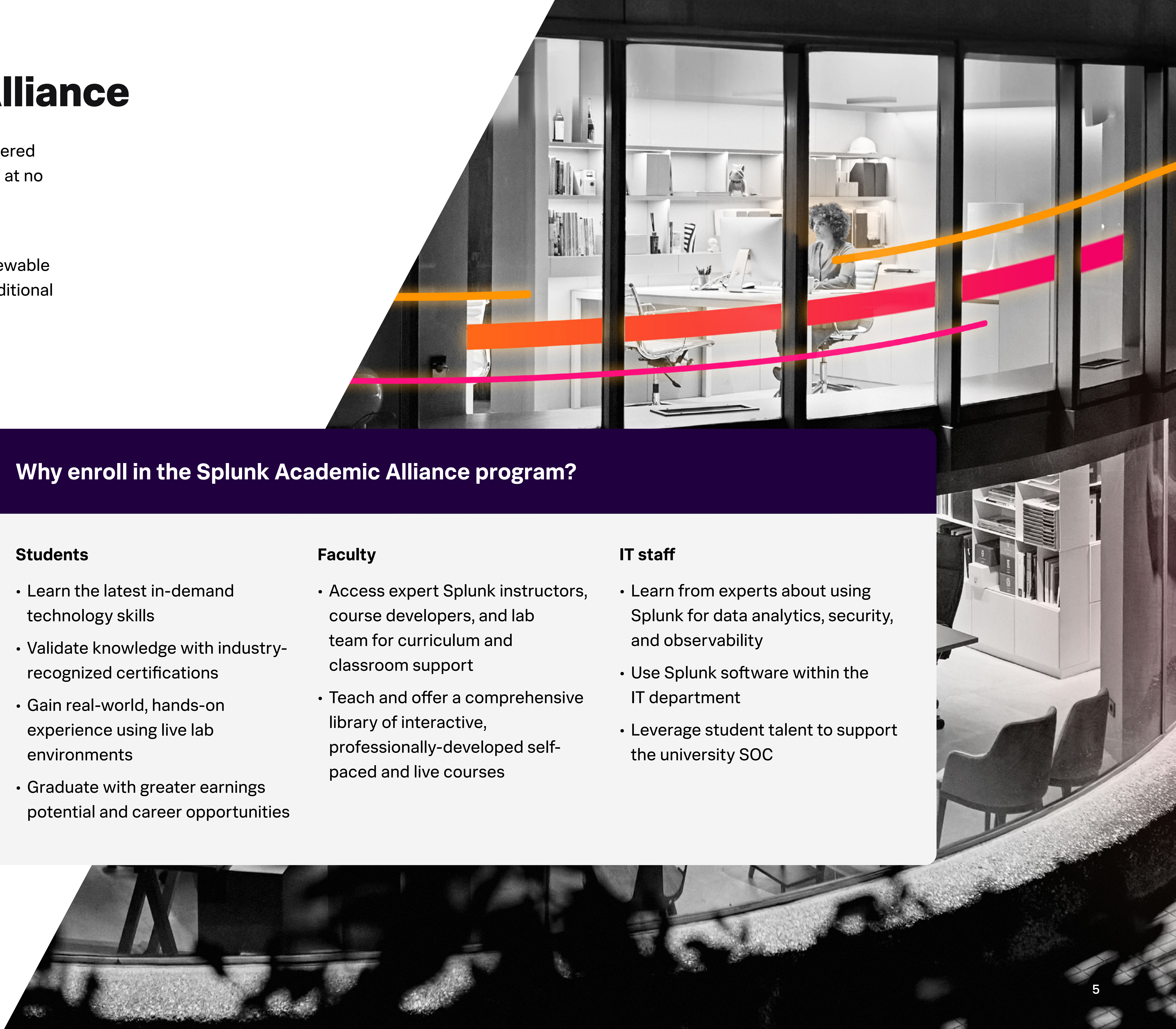
Through product donations and workforce development, we give academic institutions equitable access to our platform (Splunk Enterprise) with a renewable one-year, 10GB license for Splunk software and access to eLearning and additional program benefits for qualifying not-for-profit universities — at no cost.

Learn how UNLV hit the jackpot with student SOC and Splunk

Providing students with hands-on experience in a real SOC broadens their horizons and opportunities once they enter the workforce.

— Vito Rocco, CISO, University of Nevada, Las Vegas

100% **job placement post-graduation**



Why enroll in the Splunk Academic Alliance program?

Students

- Learn the latest in-demand technology skills
- Validate knowledge with industry-recognized certifications
- Gain real-world, hands-on experience using live lab environments
- Graduate with greater earnings potential and career opportunities

Faculty

- Access expert Splunk instructors, course developers, and lab team for curriculum and classroom support
- Teach and offer a comprehensive library of interactive, professionally-developed self-paced and live courses

IT staff

- Learn from experts about using Splunk for data analytics, security, and observability
- Use Splunk software within the IT department
- Leverage student talent to support the university SOC

Three program tiers support different need levels

We offer three program tiers so anyone attending, teaching, or working at an academic institution can learn to use Splunk and get Splunk certified for free or at a discount.

Foundational tier

This no-cost program for students, faculty and staff includes:

- 21 online, self paced learning courses
- Lab environment for real-life learning
- Certification
- Program management and curriculum via Splunk LMS

Intermediate tier

Building on the foundational tier, this no-cost program for students and faculty includes:

- Program workshops or bootcamps
- Additional e-learnings for Enterprise Security (ES) and Security Orchestration, Automation and Response (SOAR)
- Flexible scheduling of live instruction and hands-on labs
- Train-the-trainer options
- Program design, delivery, and support through Splunk Education team

Advanced tier

This takes the foundational and intermediate tiers further at no cost to students and faculty. It includes:

- Integrated customized program curriculum
- Live instruction and hands-on labs
- Flexible scheduling of live instruction and hands-on labs
- Program design and support for SOC models and cyber analyst challenges

Building a student-powered SOC: Roles and responsibilities

As you start working with Splunk through our Academic Alliance program, you'll want to think through how you'll start integrating student staffers into your SOC. All tier 1 analysts will most likely begin by handling tier 1 incident triage and response. As your tier 1 analysts progress in skill, some of your student staffers will move into a tier 2 role, and will be involved in incident response.

Our engineers recommend, in addition to the training plans on the following page, a few workshops to supplement your students' learning. These can include a Security Lunch n Learn, Enterprise Security Hands-on, and Investigating with Splunk.

Building a student-powered SOC: Suggested training plans

Student SOC analyst 1 path

Practical Application

- Cybersecurity Defense Analyst Certificate
- Enterprise Security Hands-on
- Risk-Based Alerting Hands-on
- UBA Hands-on
- SOAR Hands-on
- Security Operations Suite Hands-on

Student SOC analyst 2 path

(In addition to the above analyst 1 path)

Advanced Analysis

- Investigating with Splunk
- Splunking the Endpoint
- Advanced APT Hunting
- AWS Hands-on / AWS2: Attack in the Clouds

Content Development

- Creating Dashboards with Splunk
- Search Optimization
- Building Correlation Searches
- Developing SOAR Playbooks

Student SOC Splunk admins and engineers path

Practical Application

- Splunk Cloud Administration / Transitioning to Splunk Cloud
- Splunk Enterprise Data Administration
- Administering Splunk Enterprise Security
- Advanced SOAR Implementation
- Admin Dispatches from the SOC

Advanced Administration

- Architecting Splunk Enterprise Deployments
- Splunk Enterprise Cluster Administration
- Troubleshooting Splunk Enterprise
- Splunk Enterprise Practical Lab

Three bellweathers of the student-powered SOC

Institutions of higher learning vary wildly in size, geographical location(s), and mission. As no two institutions are the same, the student-powered SOC model can't be, either. Here are three examples of institutions that have built student-powered SOCs through the Splunk Academic Alliance.

California Polytechnic State University

The Cal Poly "junior security operations center analyst" is expected to work between 10–16 hours per week during the school year. The so-called "learning SOC" provides the students an opportunity to learn real-world tools, like their go-to SIEM product, Splunk Enterprise Security, and to address questions as they arise. They typically have 2-3 students working in the SOC.

"We do lots of over-the-shoulder training with them," Lomsdalen said. "We walk them through how to address specific alerts and emails, and then we give them the keyboard and go from there. They're always being monitored by full-time staff — our students are never working by themselves."

University of Cincinnati

UC's SOC program now relies on both full-time employees and about four to seven graduates at any given time. The university plans to increase the level of student participation in the future. The University of Cincinnati has many international students, and the SOC is able to give them the chance to gain real-world experience performing security operations. To accomplish that goal UC segregates some sensitive data before it is put into Splunk to allow international students to get real-world experience and internship credit in cybersecurity, since most US corporations restrict foreign students from these types of work experiences.

Louisiana State University

"Students are trained by our partner TekStream and by LSU, doing remediation detection and response alongside TekStream. It will be a true real-world experience," said Craig Woolley, CIO at LSU. Woolley added that if students are unable to handle an incoming incident, TekStream will handle it. A team of students staff the SOC Monday-Friday from 8 a.m. until 8 p.m., with TekStream providing 24/7 support and monitoring during those times and also during the overnight hours.

LSU will operate two student-powered SOCs at its Baton Rouge and Shreveport campuses, and eventually, the SOC capabilities will be scaled to support any institution across the state wishing to develop one.

This program has expanded into the State of Louisiana agencies, they are offering students who come out of this program state employee jobs that come with student loan forgiveness if they fulfill a two year commitment. This means of keeping cybersecurity talent in-state is gaining traction, and other states are in the process of implementing similar programs.

The key to implementation success: Splunk's partner community

Splunk's partner community plays a vital role in the successful implementation of our student-powered SOCs. There are variations in how they support these efforts but there are some common threads.

Our managed security service providers (MSSPs) provide important coverage and act as a training partner for students, often providing core 24x7 monitoring, threat detection, alerting, and security monitoring. As students are onboarded into the program, they are integrated into tier 1 of the SOC. They will typically cover daylight hours alert triage, and the MSSP will then cover the overnight monitoring and response. As the students' abilities and interests grow, some of them will move into tier 2 of the SOC and begin to do incident response handling or other more sophisticated duties.

In some cases, Splunk partners even take on time-consuming administrative duties, tracking students' progress for their work study credits and industry certifications.

Getting started with the Academic Alliance Program

If you are a faculty member or university professor* who wants to use Splunk to teach students how to leverage big data analytics, practical cybersecurity skills and data-driven research insights, and meet the program requirements, then you're ready to apply.

Program Requirements

- Apply through Splunk's Global Impact Program
- Be a US-based 501c3 or
- Certified international charity or
- Public or private nonprofit institution of higher education

* Please note that the enrollment form must be completed by a college/university administrator or faculty member — students cannot enroll themselves.

Enroll now

25_CMP_ebook_the-student-powered-SOC-handbook_v8